

Chapter 2. Examples of distribution laws of random variables

http://www.simumath.com/library/book.html?code=Mat_Stat_random_values

2.1 Density function and probability distribution function

Variable is called *random* if as result of experience it can accept valid values with certain probabilities.

Law of distribution is function (given by table, graph or formula), allowing to define probability of that random variable

A random variable can be discrete

$$P\{X = x_i\} = p_i, \quad i = 1, 2, 3, \dots \quad \sum_{i=1}^{\infty} p_i = 1, \quad (1)$$

The sum of all probabilities is one.

**A random variable can be continuous
if the probability of falling into the interval (a, b)**

$$P\{a \leq X \leq b\} = \int_a^b f(x) dx. \quad (2)$$

**Function $f(x)$ is called *density function*
of continuous random variable.**

$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad \text{and} \quad f(x) \geq 0$$

The sum of all probabilities is one.

Probability of random variable X (discrete or continuous) accepts value, smaller x , is called ***distribution function*** of random variable X and is designated **$F(x)$** :

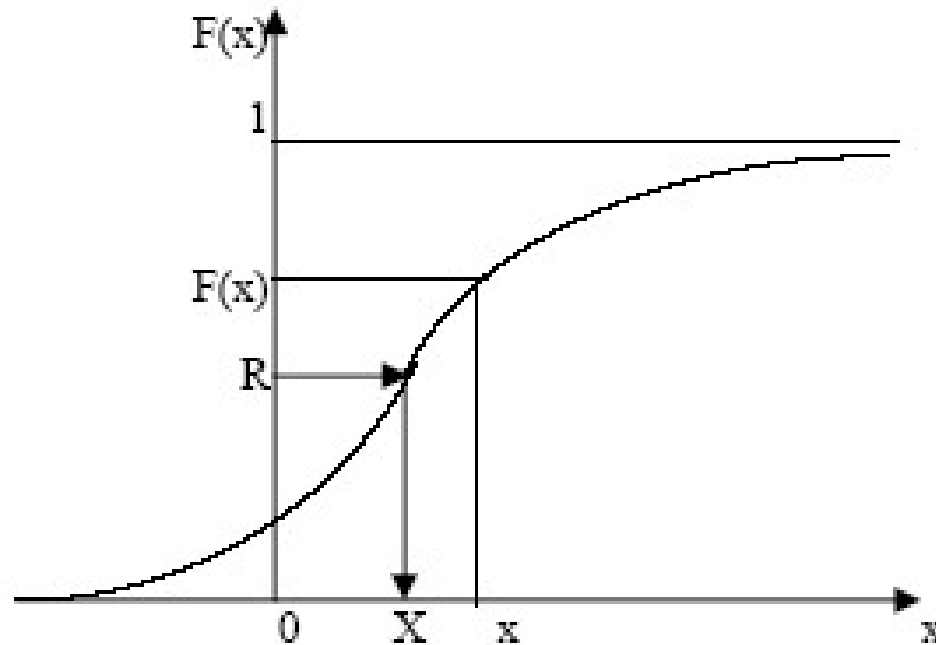
$$F(x) = P\{X < x\} \quad \text{distribution function}$$

For a continuous random variable, the probability distribution function is

$$F(x) = P\{X < x\} = \int_{-\infty}^x f(x) dx.$$

Integral from minus infinity to x

Typical graph of a probability distribution function is



Basic properties of distribution function:

- 1) $F(x)$ – not decreasing function, i.e. at $x_2 > x_1$ $F(x_2) \geq F(x_1)$,
- 2) $F(x)$ – limited function: $0 \leq F(x) \leq 1$,
- 3) $\lim_{x \rightarrow -\infty} F(x) = 0$,
- 4) $\lim_{x \rightarrow +\infty} F(x) = 1$,

not less than zero, but not more than one

2.2 Characteristics of the distribution of a random variable

Mathematical expectation (mean or expected value $M(X) = E(X) = \mu_X$ of discrete random variable X is the sum:

$$M(X) = E(X) = \mu_X = x_1 \cdot p_1 + x_2 \cdot p_2 + x_3 \cdot p_3 + \dots + x_n \cdot p_n = \sum_{i=1}^n x_i p_i \quad 6(a)$$

Mathematical expectation of continuous random variable X is the integral from product of its values x and its **density function $f(x)$** :

$$M(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx. \quad (6b)$$

Mathematical expectation characterizes *average value* of random variable X and is the distribution center

Dispersion. Dispersion of random variable X is the number:

$$D(X) = M \{ [X - M(X)]^2 \} = M(X^2) - [M(X)]^2. \quad (8)$$

Proceeding from definitions of dispersion (8) and mathematical expectation (6a) for discrete random variable and (6b) for continuous random variable we'll receive:

$$D(X) = M(X - M) = \sum_{i=1}^n (X_i - m)^2 \cdot P_i \quad (9) \text{ for discrete random variables.}$$

$$D(X) = M(X - M) = \int_{-\infty}^{+\infty} (x - m)^2 \cdot f(x) \cdot dx \quad (9) \text{ for continuous random variables}$$

Here $m = M(X)$.

$$\textbf{Standard deviation: } \sigma = \sqrt{D(X)}. \quad (11)$$

Standard deviation characterizes the **width** of the probability distribution of a random variable.

Median of random variable X is its value Me for which the equality

$$P\{X < Me\} = P\{X > Me\}$$
 takes place, i.e. equiprobably

In case of symmetric distribution median and mathematical expectation coincide.

2.3 Examples of distribution laws

Binomial distribution. Discrete random variable X has *binomial distribution*, if its possible values $0, 1, 2, \dots, m, \dots, n$, and probabilities corresponding them are equal to

$$P_m = P(X = m) = C_n^m \cdot p^m \cdot q^{n-m}, \quad (21)$$

where $0 < p < 1$, $q = 1 - p$; $m = 0, 1, 2, \dots, n$.

Here $C_n^m = \frac{n!}{m!(n-m)!}$ is the binomial coefficient,

$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$ is a factorial. For instance $4! = 1 \cdot 2 \cdot 3 \cdot 4 = 24$

Mathematical expectation: $E(X) = \mu_X = n \cdot p$

Dispersion: $D(X) = n \cdot p \cdot q$

Standard deviation: $\sigma = \sqrt{n \cdot p \cdot q}$

Example 2.1

Construct binomial distribution for $n=2$ and $p=0.5$.

According to the equation (21)

$$m = 0, 1, 2; \quad q=1-0.5=0.5.$$

$$P_0 = \frac{2!}{0!(2-0)!} 0.5^0 \cdot 0.5^{2-0} = 0.25$$

$$P_1 = \frac{2!}{1!(2-1)!} 0.5^1 \cdot 0.5^{2-1} = 2 \cdot 0.25 = 0.5$$

$$P_2 = \frac{2!}{2!(2-2)!} 0.5^2 \cdot 0.5^{2-2} = 0.25$$

In this way

m	0	1	2
P_m	0.25	0.5	0.25

The sum of all probabilities is
 $0.25 + 0.5 + 0.25 = 1$.
Distribution is valid.

Example 2.2

Construct binomial distribution for $n=3$ and $p=0.5$.

Find mean or expected value $E(X) = \mu_X$

Find variance (dispersion) and standard deviation.

Example 2.3

Construct binomial distribution for $n=4$ and $p=0.5$.

Find mean or expected value $E(X) = \mu_X$

.Find variance (dispersion) and standard deviation.

Example 2.4

With MAPLE construct binomial distribution for $n=10$ and $p=0.25$; .

$p=0.5$;

$p=0.75$.

Construct histogram.

Find mean or expected value $E(X) = \mu_X$

Find variance (dispersion) and standard deviation.

Example 2.5

With MAPLE construct binomial distribution for $n=15$ and $p=0.25$; .

$p=0.5$;

$p=0.75$.

Construct histogram.

Find mean or expected value $E(X) = \mu_X$

Find variance (dispersion) and standard deviation.